

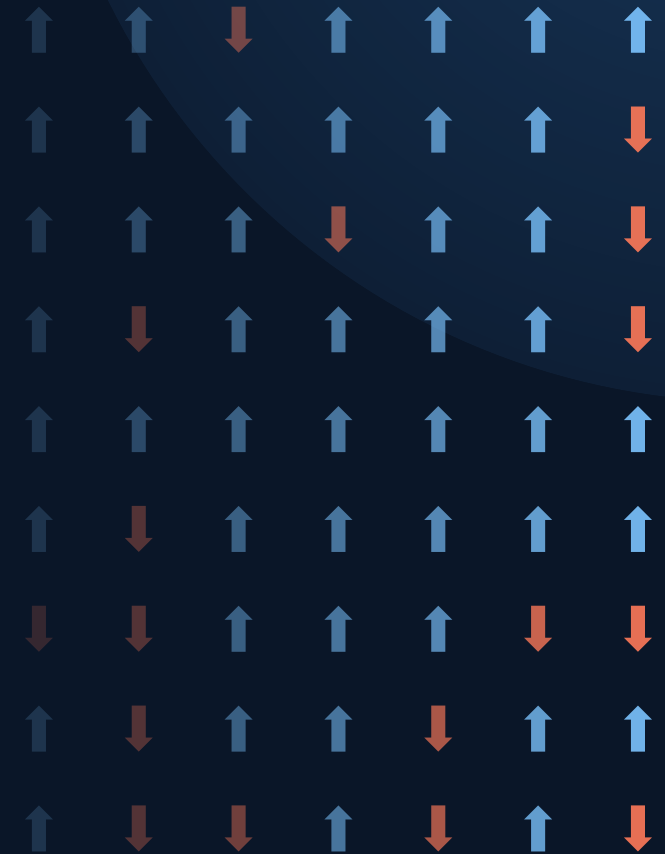
Measuring the External Field

Can we measure how hard a news article pushes on what people think?

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each arrow is one person's opinion



Physics treats media as a push, h , on opinion. Nobody measures h

$$m = \tanh(\beta J m + h)$$

 m

average opinion
-1 all against · +1 all for

 J

pull toward
the people around you

 β

how firmly people
follow that pull

 h

media's push,
the same on everyone



Question: can h be measured from the content itself?



Testing the tools changed five assumptions

WE PLANNED

❌ Measure the amygdala

❌ Score as a ratio

❌ Outrage stands out most

❌ Measure α , media's strength

❌ A manipulation detector

WE FOUND

✅ Model sees brain surface only, so we used surface areas

✅ Ratio broke on 69 of 400, so we used a difference

✅ Fear stands out; outrage lifts both areas equally

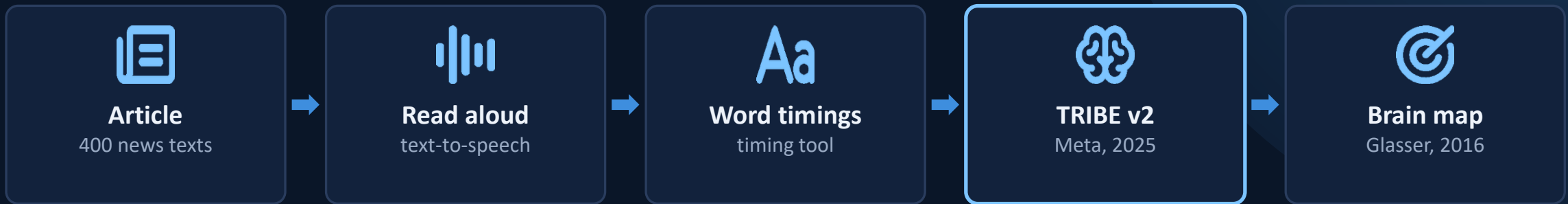
✅ Data can't pin α , so we derived its minimum

✅ Word counting wins, so it's a measurement

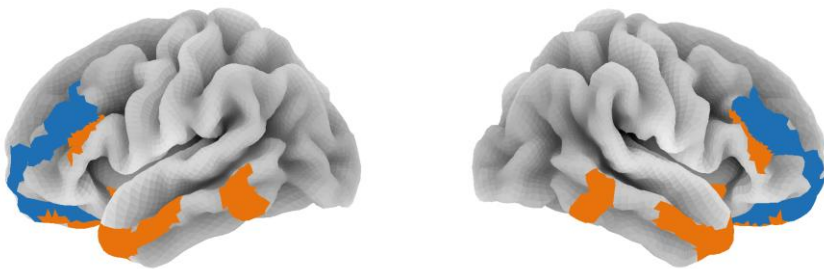
All five are in the approved amendment.



How one article becomes one number



ROI definition, not a measurement: affective-salience (1030 vertices, orange) and deliberative-control (851, blue)
left right



orange 1,030 emotion points blue 851 reasoning points

THE SCORE

$$X = \bar{A}_{\text{emotion}} - \bar{A}_{\text{reasoning}}$$



Nobody scanned: every value is predicted



400 articles, four groups, matched in length

100

Fear-driven

fake news (ISOT)

100

Outrage

partisan news (SemEval)

100

Clickbait

headlines (Webis)

100

Neutral

medical + real news

≈ 164

words per article,
all groups within 5 words

0.027

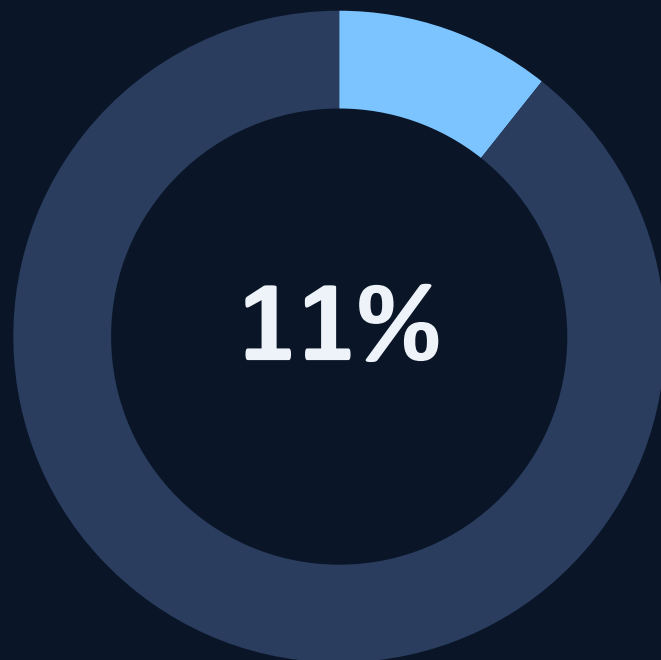
smallest effect we
could detect, set in advance

× 2

every article
scanned twice



Result 1: the four groups really differ



11% of the differences between articles comes from their group

$$\eta^2 = \frac{\text{between groups}}{\text{total}} = 0.107$$

η^2 ON EACH RUN, AGAINST THE SMALLEST WE COULD DETECT



1 in a billion

chance it is luck ($p = 1 \times 10^{-9}$)

Repeats

second run agrees (0.089)

Meaning: the groups differ for real, but most variation is from one article to the next.

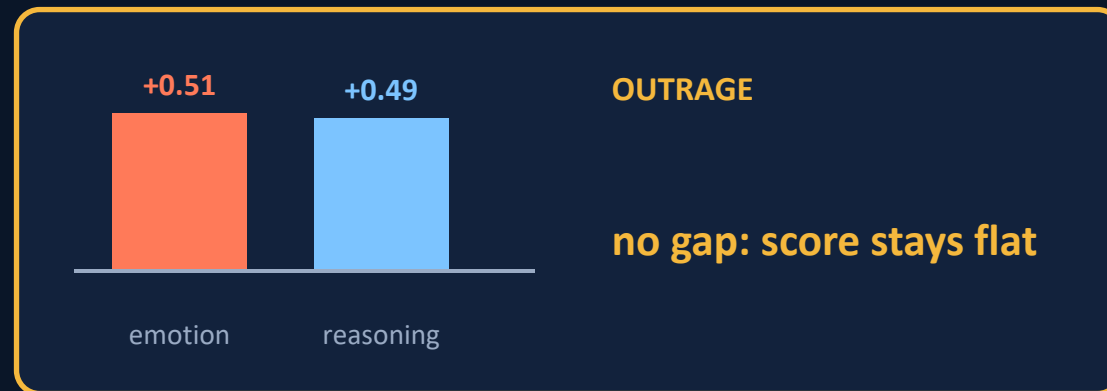
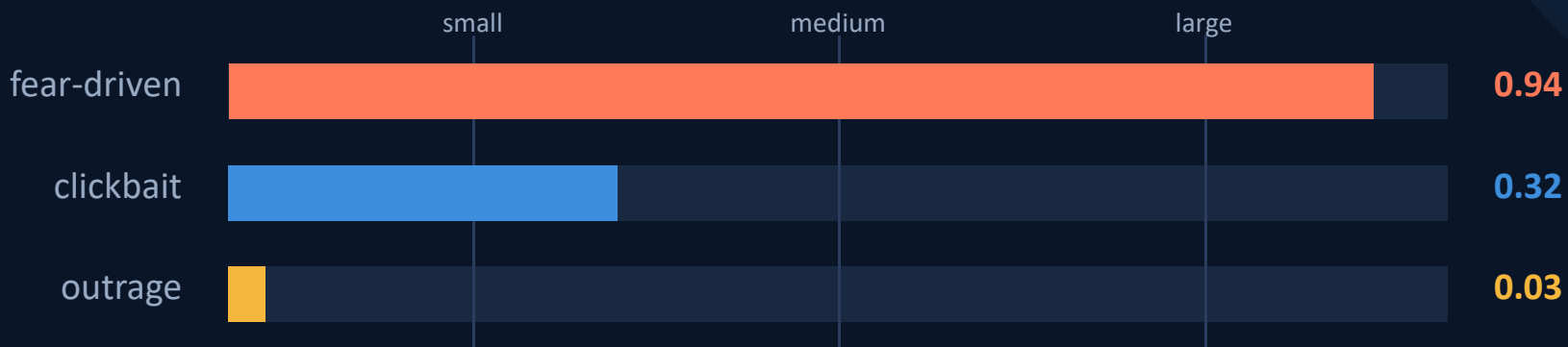


THE RESULTS

Result 2: fear drives it; outrage lifts both sides equally

$$d = \frac{\bar{X}_{\text{group}} - \bar{X}_{\text{neutral}}}{s}$$

distance from neutral, in spreads (s)





Result 3: scores repeat for groups, not single articles

0.87

agreement between two runs (ICC)



0 none

1 identical

51 of 400

flipped sign between runs



• saw 51 | noise predicts 55 (shaded 44 to 67)

Meaning: trust group averages. Never judge one article: its score sits near zero and noise can tip it.

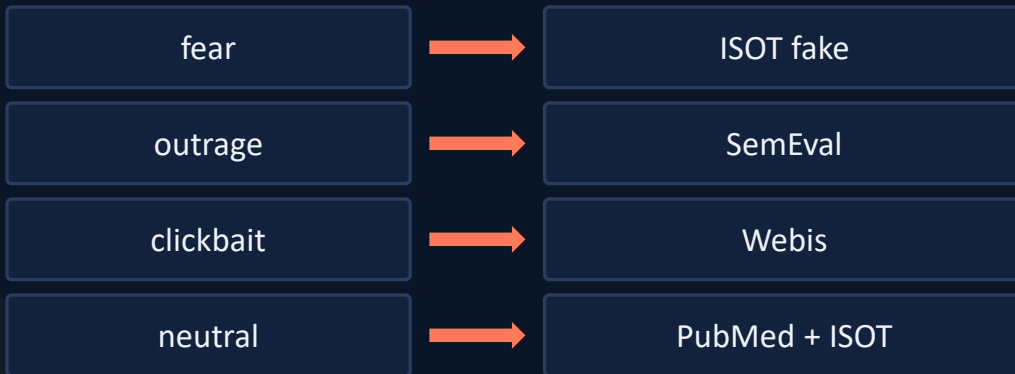


Honest check: word counting beats our score

How well each method sorts manipulative from neutral articles (AUC)



WHY: EACH GROUP CAME FROM ONE SOURCE



66%

words guess the source
(chance: 25%)

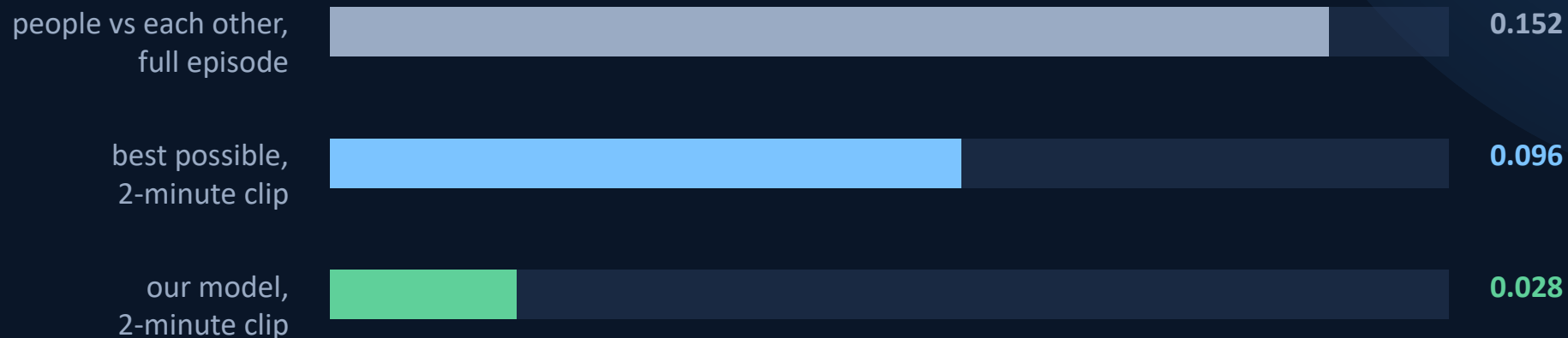
So our score is a
measurement for the
physics, not a detector.



THE RESULTS

Does the brain model match real brains? Weakly, yes

Tested on public brain scans of people watching Friends (Algonauts 2025). Higher = closer match (r).



≈ 30%

of the best possible on the clip

$p = 0.048$

just under the 5% cut-off

3 bugs

found and fixed along the way

Meaning: weak but positive, not the opposite of real brains. A longer run will settle it (Paper 3).



The minimum push media would need

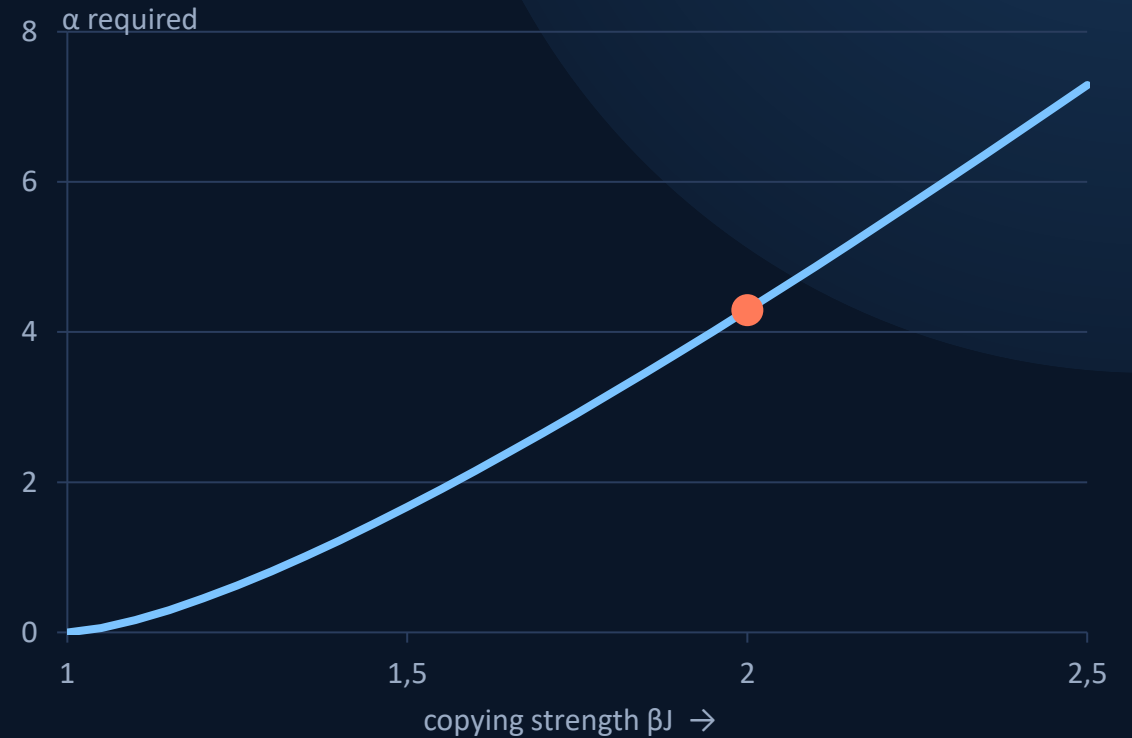
$$h = \alpha X \Rightarrow \alpha \geq \frac{h_c(\beta J)}{\Delta X}$$

α strength of the push per unit of score

h_c smallest push that flips the majority

βJ how strongly people copy each other

ΔX spread of our scores = 0.124



$\alpha \geq 4.29$ at $\beta J = 2$
a bar any claim must clear



Can it analyse any content, even what children watch?



CAN DO

Score video, audio and text

Compare groups of content

Rate content, never the viewer



NOT YET

Model learned from adult brains

Only news text was tested

One show alone is unreliable



WOULD TAKE

Child brain data, with ethics approval

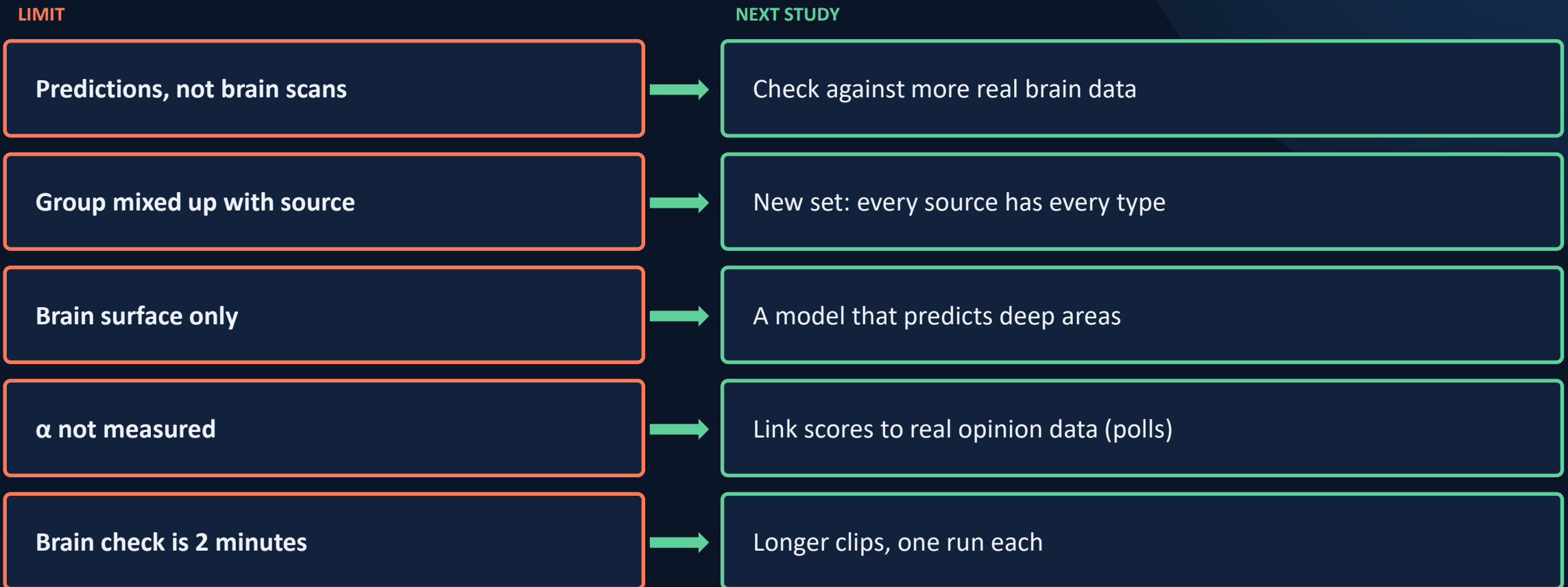
A children's video test set

Safeguards against targeting children

Short answer: technically possible, but not yet valid for children.



Each limit points to the next study





Taking it further: where this research can go



PUBLISH

Paper 1 → Physica A

Paper 2 → Physica A

Paper 3 → Imaging Neuroscience

Preprints first (arXiv)



EXTEND THE SCIENCE

Kenya: 2024 Finance Bill coverage

Video and audio content

Measure α with opinion data

Children's media, ethically



OPPORTUNITIES

Postgraduate research topic

Neuroscience + media collaborations

Open tool, already online

Group-level media audits

Live instrument: monarch-4iy.pages.dev · Code and data: github.com/brn-mwai/monarch



Three papers, and what I need from you today

PAPER 1 · Physica A

The minimum push

12 pp · ready

PAPER 2 · Physica A

The measuring tool

16 pp · ready

PAPER 3 · Imaging Neurosci.

Matching real brains

8 pp · one longer run

TODAY I AM ASKING FOR



1 Any corrections



2 Signature, page iii



3 Library clearance by Thu
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4 Co-authorship view